

Lipids

The term lipids are refers to a collection of organic molecules of varying chemical compositions present in plant and animal tissues, they are grouped together on the basis of their solubility in non-polar solvents such as acetone, chloroform, and benzene.

Biomedical importance of lipids:-

1. Act as energy source and storage in the body.
2. Lipids supply essential fatty acids in diet.
3. They serve as a source of fat soluble vitamins (Vitamin A, D, E and K).
4. Lipids act as protective layer for the vital organs.

5. Lipids act as insulator because fat stored beneath the skin (subcutaneous fat).
6. Lipids include a number of hormones, of which the most important are steroids and prostaglandins.
7. Phospholipids are important constituents of cell membranes.
8. Lipoproteins are carriers of Triglycerides, phospholipids and cholesterol in the blood.

Lipids are commonly subdivided to four main groups:

1. Fatty acids (saturated and Unsaturated).
2. Glycerides (glycerol containing lipid).
3. Nonglyceride lipids (sphingolipids, steroids and waxes).
4. Complex lipid (lipoproteins).

1- Fatty acids:

Fatty acids are aliphatic (also called an **acyl chain**) carboxylic acids and have the general formula, $R\text{—COOH}$

Fatty acids in most biological systems are synthesized by serial addition of two carbon units, to produce an even number carbon atom fatty acid.

Over 100 fatty acids have been isolated from various lipids.

The naturally occurring fatty acids are carboxylic acids with un-branched hydrocarbon chains.

They are present in all organisms as components of fats and membrane lipids. In these compounds, they are esterified with alcohols (glycerol, sphingosine, or cholesterol).

However, fatty acids are also found in small amounts in unesterified form. In this case, they are known as free fatty acids (FFAs).

Depending on total number of carbon atoms

a. Even chain

They have carbon atoms 2,4,6 and similar series. Most of the naturally occurring lipids contain even chain fatty acids.

a. Odd chain

They have carbon atoms 3, 5, 7, etc. Odd numbered fatty acids are seen in microbial cell walls. They are also present in milk.

Depending on length of hydrocarbon chain fatty acid divided to:

- a. Short chain with 2 to 6 carbon atoms
- b. Medium chain with 8 to 14 carbon atoms
- c. Long chain with 16 to 22 carbon atoms.
- d. Very long chain fatty acids (more than 24 carbon)

The numbering of fatty acids are numbered as C1, C2, ..etc starting from the COOH group..

Depending on the absence, or presence of double bonds, they are classified into saturated and unsaturated fatty acids.

Saturated fatty acids:-

Fatty acids that do not contain double bond in their structure are called saturated fatty acids.

Palmitic acid	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$
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Stearic acid	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$
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Δ nomenclature

The majority of fatty acids are linear molecules with carboxylic acids at one end.

The only differences between fatty acid molecules are number of carbons, the number and position of double bonds.

The symbol “ Δ ” is used to indicate the presence and position of the double bond.

The positions of any double bonds are specified by superscript numbers following the Δ .

Unsaturated fatty acids:

Monounsaturated fatty acids contain a single double bond, while others containing more than one site of unsaturation which called poly Unsaturated fatty acids.

for example:

Palmitoleic acid [16:1 Δ^9]

Oleic acid [18:1 Δ^9]

(it mean an 18 carbon fatty acid with one double bond between the C₉-C₁₀).

The first double bond added to saturated fatty acid is nearly always in the Δ^9 position by an enzyme system Δ^9 desaturase, therefore palmitoleic and oleic acids are non essential fatty acids. (Essential fatty acids are fatty acids that have to be supplied in the diet.) and only linoleic and linolenic acids are considered to be essential fatty acids.

Polyunsaturated fatty acids:

Linoleic acid $[18:2\Delta^{9.12}]$

Two double bonds between C_9 - C_{10} and
 C_{12} - C_{13} .

Linolenic acid $[18:3\Delta^{9.12.15}]$

Contain three double bonds between C_9 - C_{10} ,
 C_{12} - C_{13} and C_{15} - C_{16} .

Arachidonic acid $[20:4\Delta^{5.8.11.14}]$

Contains four double bonds between C_5 - C_6 ,
 C_8 - C_9 , C_{11} - C_{12} and C_{14} - C_{15} .

For unsaturated fatty acid there are two possible isomers cis and trans, usually only the cis forms are found in natural lipids.

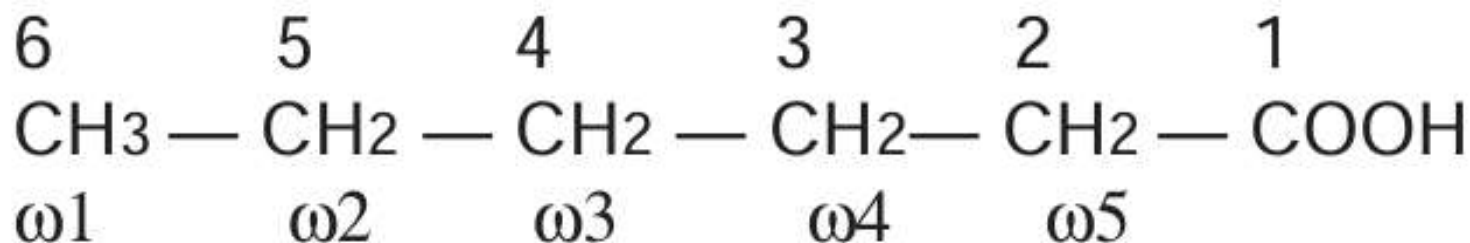
Branched fatty acids only occur in bacteria.

Omega ω nomenclature

Another numbering system that is frequently used by nutritionists is the omega system.

Omega is the last letter in the Greek alphabet.

With this system the carbon atom farthest from the carboxyl carbon (the methyl carbon) is called omega (ω); and counting back along the chain of carbon atoms gives ω_1 , ω_2 , ω_3



For example a fatty acid with a double bond between the third and fourth carbon atoms from the methyl carbon is called an ω_3 fatty acid.

The position of the first double bond is indicated after the (ω). Often in the ω system, the other double bonds are not specified.

The nomenclature has the advantage that the position numbers of the double bonds do not change with elongation since the new carbons are added to the carboxylate end.

By using this nomenclature,
oleic acid is an $\omega 9$ fatty acid, and
linoleic acid is an $\omega 6$ fatty acid containing two
double bonds while
Linolenic acid is an $\omega 3$ fatty acid containing
three double bonds.

Humans can synthesize $\omega 9$ fatty acids; plants
and some animals (especially fish) can
synthesize $\omega 6$ and $\omega 3$ fatty acids as well.

Essential fatty acids

Two fatty acids are dietary essentials in humans because of our inability to synthesize them: **linoleic acid**, which is the precursor of ω -6 arachidonic acid, the substrate for prostaglandin synthesis, and α -linolenic acid, the precursor of other ω -3 fatty acids important for growth and development.

[Note: Arachidonic acid becomes essential if linoleic acid is deficient in the diet.]

Many clinical studies have shown positive roles for omega-3 fatty acids in cancer; cardiovascular diseases; and more recently, in various mental illnesses. These fatty acids are known to have effects against inflammation, platelet aggregation, hypertension, and hyperlipidemia.

These beneficial effects may be mediated through several distinct mechanisms, including alterations in cell membrane composition and function, gene expression, or eicosanoid production.

Table 26.1 The Most Abundant Fatty Acids in Animal Fats, Vegetable Oils, and Biological Membranes

Carbon Atoms/ Double Bonds*	Structure	Common Name	Melting Point (°C)
Saturated Fatty Acids			
12:0	$\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$	Lauric acid	44
14:0	$\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$	Myristic acid	58
16:0	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$	Palmitic acid	63
18:0	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$	Stearic acid	70
20:0	$\text{CH}_3(\text{CH}_2)_{18}\text{COOH}$	Arachidic acid	77
Unsaturated Fatty Acids			
16:1	$\text{CH}_3(\text{CH}_2)_5\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	Palmitoleic acid	1
18:1	$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	Oleic acid	16
18:2	$\text{CH}_3(\text{CH}_2)_4(\text{CH}=\text{CHCH}_2)_2(\text{CH}_2)_6\text{COOH}$	Linoleic acid	-5
18:3	$\text{CH}_3\text{CH}_2(\text{CH}=\text{CHCH}_2)_3(\text{CH}_2)_6\text{COOH}$	Linolenic acid	-11
20:4	$\text{CH}_3(\text{CH}_2)_4(\text{CH}=\text{CHCH}_2)_4(\text{CH}_2)_2\text{COOH}$	Arachidonic acid	-49

*The first number is the number of carbons in the fatty acid; the second is the number of carbon-carbon double bonds in its hydrocarbon chain.

Melting point of fatty acid:

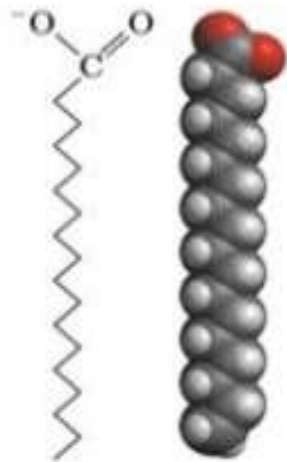
Melting points are strongly influenced by the length and degree of unsaturation of the hydrocarbon chain. At room temperature, the saturated fatty acids from 12:0 to 24:0 have a waxy consistency, whereas unsaturated fatty acids of these lengths are oily liquids.

This difference in melting points is due to different degrees of packing of the fatty acid molecules. In the fully saturated compounds, free rotation around each carbon–carbon bond gives the hydrocarbon chain great flexibility.

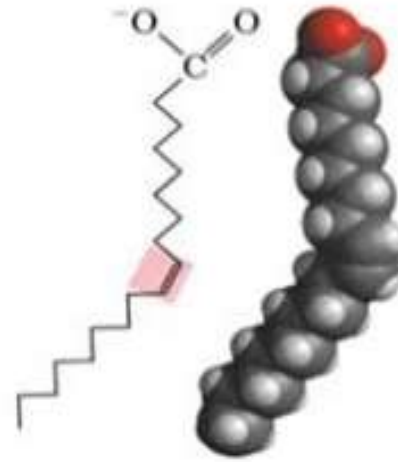
Fatty Acid Structure

Carboxyl group

Hydrocarbon chain



(a)

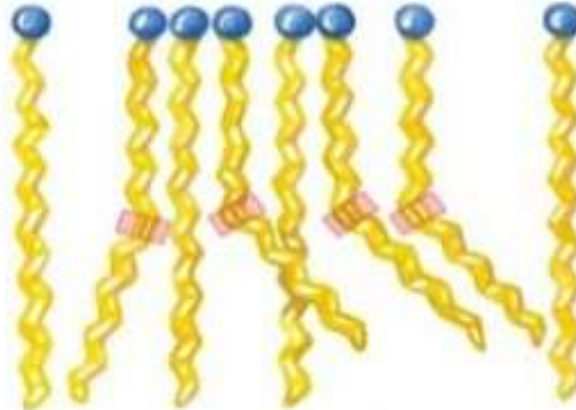


(b)



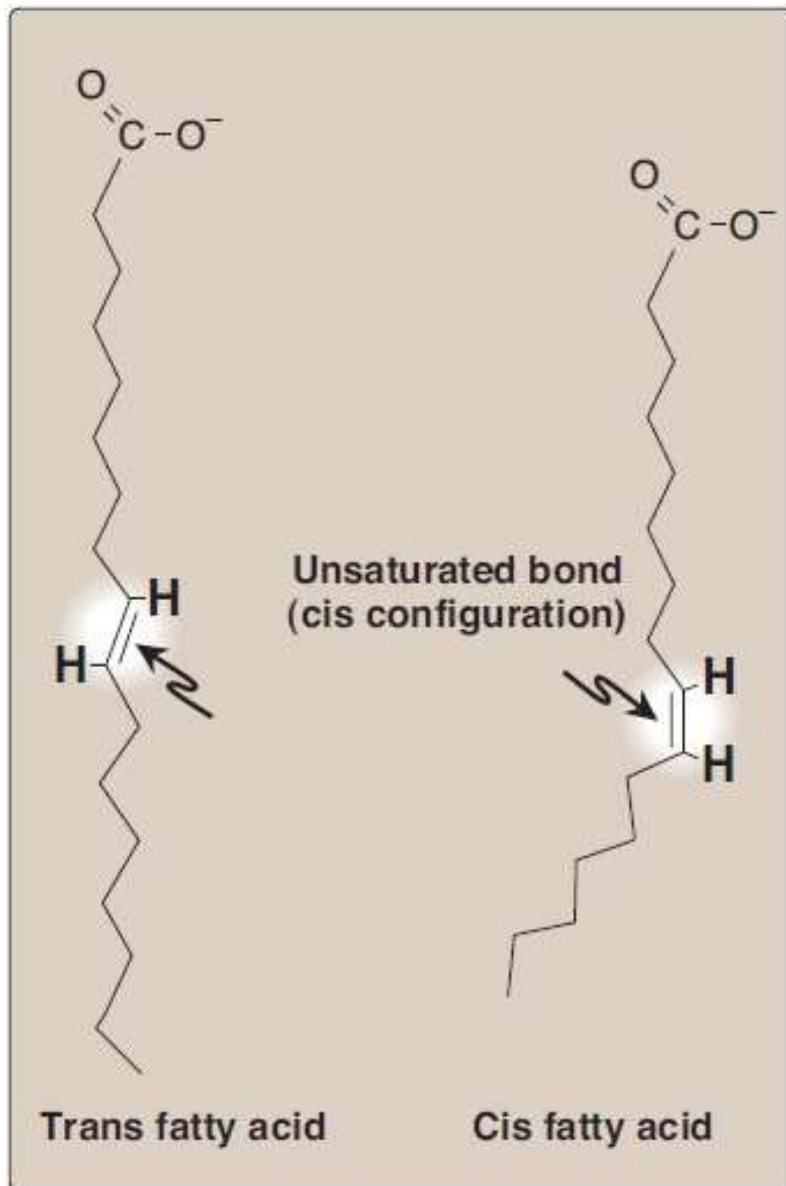
Saturated fatty acids

(c)



Mixture of saturated and unsaturated fatty acids

(d)



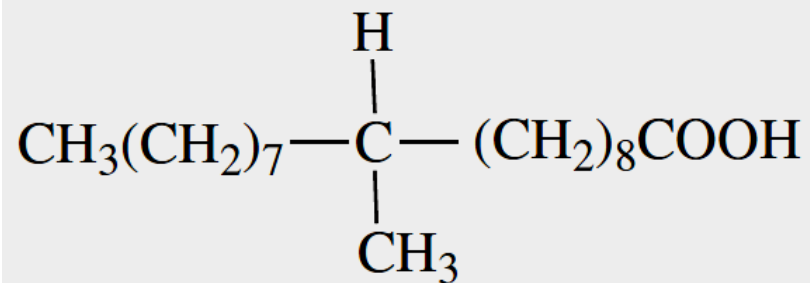
These molecules can pack together tightly in nearly crystalline arrays, with atoms all along their lengths in van der Waals contact with the atoms of neighboring molecules. In unsaturated fatty acids, a cis double bond forces a kink in the hydrocarbon chain. This decreases van der Waals contacts: the presence of cis double bonds therefore results in lower melting temperature because they result in less regular and less stable structures.

The melting points of different fatty acids differ markedly as shown in this table

Saturated Fatty Acids	
Number of Carbons	Melting point (°C)
10	32
12	44
14	54
16	63
18	70
20	77
22	82
24	86

Differences in melting temperature exist between fatty acids containing the same number of carbon atoms because the preferred conformation of a chain of saturated carbon atoms is a long, straight structure. A trans double bond does not cause a bend in the chain, but a cis double bond causes a bend in the structure, straight molecules can pack together more densely and gives a higher melting temperature.

Branching can also affect melting temperature; for example: 10-methyl stearic acid melts at only 10°C



Longer chains result in higher melting temperatures; increasing numbers of double bonds decreases melting temperature for fatty acids of a given length.

Effect of Unsaturation on Fatty Acids

Number of Carbons	Number of double bonds	Melting point (°C)
18	0	70
18	1	13
18	2	-5
18	3	-11

Table 3-2 Melting Points of Some Biologically Important Fatty Acids

Fatty Acid	Melting Temperature (°C)
Palmitic acid	63
Stearic acid	70
Oleic acid	13
Elaidic acid (<i>trans</i> -9-octadecanoic acid)	44
Linoleic acid	-5
α -Linolenic acid	-11

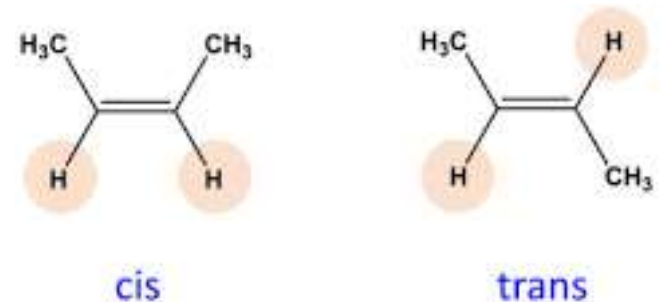
Trans Fatty Acids or Trans Fats

Trans fatty acids are principally artificial fats but a small amount of TFAs occur naturally in meat and dairy products.

Trans fatty acids are formed artificially during the hydrogenation of vegetable oils. In this process, the vegetable oils which are liquid turn to solid.

Oils containing poly unsaturated FA also have high content of Trans FA.

Fast food preparations have a high Trans FA content.



Hydrogenation improves the shelf-life and palatability of oils and thus they are used in food industry.

Effects of Trans Fatty Acids in our body:

TFAs are more atherogenic than saturated fatty acids.

- Increasing LDL levels (bad cholesterol)
- Decreasing HDL (good cholesterol) level
- Increasing the abnormal clotting of blood.
- insulin resistance and diabetes.

Eicosanoids (*eikos* meaning twenty):

Some of the unsaturated fatty acids contain more than one double bond such as linoleic and linolenic acid which required for the biosynthesis of arachidonic acid are the precursor of a class of hormone like molecule known as eicosanoids (prostaglandin, leukotriene and thromboxane).

They are paracrine hormone like molecules and differ from endocrine hormones in that they are not transported between tissues in the blood, but act only on cells near the point of their synthesis.

Various eicosanoids are produced in different cell types by different synthetic pathways and have different target cells and biological actions in vertebrate animals.

Eicosanoids are known to be involved in:

1. Inflammation.
2. Fever and pain associated with injury or disease.
3. Formation of blood clot.
4. Regulation of blood pressure.
5. Gastric acid secretion.

Prostaglandins (PG):

Prostaglandins are produced in low amounts in most cells in the body, including liver, kidneys, pancreas, heart, lungs, brain, intestine and seminal fluid.

The variety of prostaglandins is formed from the polyunsaturated fatty acid (Arachidonic acid) with 20 carbon structure.

Prostaglandins are further divided into two groups, ether-soluble prostaglandins (PGE), and phosphate buffer-soluble prostaglandins (PGF).

The E-type of PG have a keto group in position 9, whereas the F-type PG have a hydroxyl group in this position.

Physiologic effects of prostaglandins:-

1. Some PG regulate blood pressure.

2. Regulate body temperature.
3. Stimulate contraction and relaxation in the smooth muscle of the uterus.
4. Stimulate steroid production by adrenal glands and stimulate the release of insulin from the pancreas.
5. Stimulate the movement of calcium ions from bone.
6. PG have also used to induce abortion.
7. Producing fever, pain and cause inflammation.

When tissues are injured, arachidonic acid is converted to PG that produce inflammation and pain in the area.

The treatment of pain, fever and Inflammation (such as aspirin NSAIDs) is based on inhibiting the enzyme cyclo-oxygenase that convert arachidonic acid to PG, so decrease pain, inflammation and reduce fever.

Mechanism of Aspirin and other Pain Killers action:

Prostaglandins induce inflammation, pain, and fever.

Aspirin blocks cyclooxygenase which converting arachidonic acid to prostaglandins. The acetyl group on aspirin is hydrolyzed and then bonded to the alcohol group of serine (530) as an ester. This has the effect of blocking the channel in the enzyme and arachidonic can not enter the active site of the enzyme.

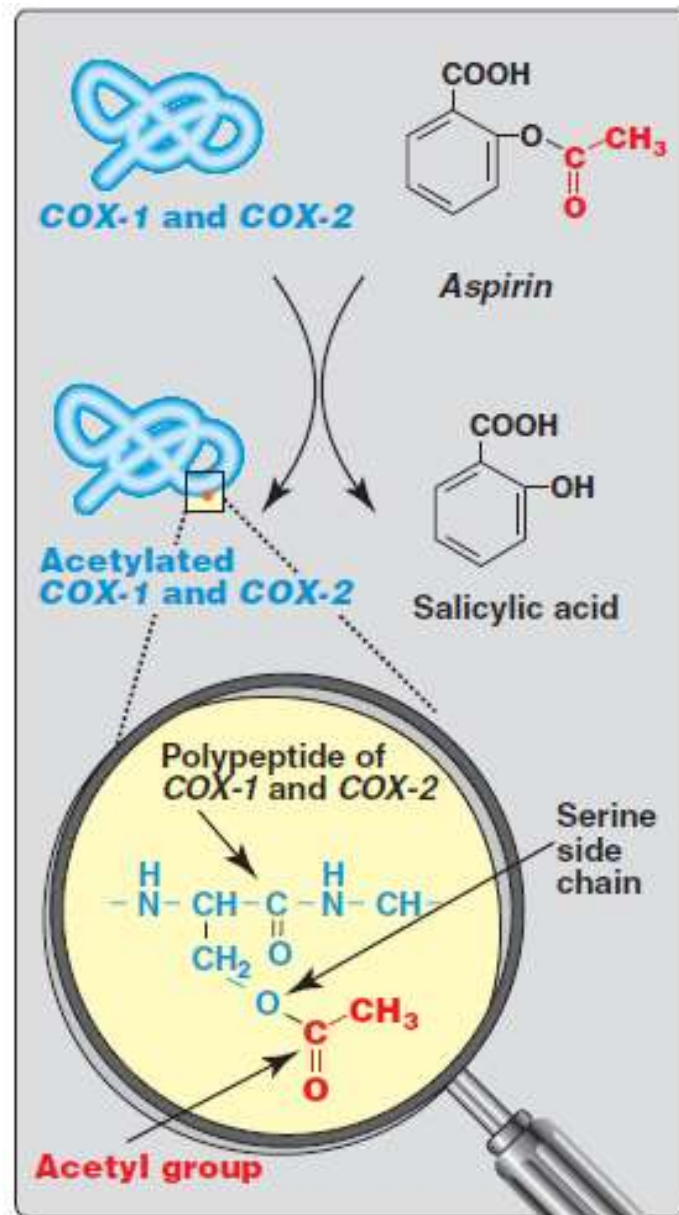


Figure 17.24
Irreversible acetylation of *COX-1*
and *COX-2* by aspirin.

By inhibiting or blocking this enzyme, the synthesis of prostaglandins is blocked, which in turn relieves some of the effects of pain and fever.

Consequently thromboxane synthesis is also inhibited, and blood clotting is impaired. This effect has prompted many physicians to recommend a daily dose of 81 mg aspirin for people at risk for heart attack or stroke.

It also explains why physicians forbid patients to use aspirin and other anti-inflammatory agents for a week before a planned surgery, aspirin and other NSAIDs may cause excessive bleeding.

Aspirin competes with arachidonate for binding to the cyclooxygenase active site. Although arachidonate binds about 10 000 times better than aspirin, once binding of aspirin acetylates a serine (530) residue at the active site to cause irreversible COX inactivation.

Although the other non-steroidal anti-inflammatory drugs inhibit COX activity, most of them cause reversible enzyme inhibition by competing with arachidonate for binding. A well-known example of a reversible NSAID is ibuprofen.

Cyclooxygenase has two isozymes.

Isoenzymes (or isozymes) are multiple form of an enzyme which differ in chemical and physical properties and catalyze the same reaction as an enzyme

Aspirin inhibit both COX-1 and COX-2. This explains why aspirin and the other anti-inflammatory agents have undesirable side effects.

NSAIDs also interfere with the COX-1 isoform of the enzyme, which is needed for normal physiological function. Their side effects include stomach and duodenal ulceration.

Thromboxanes:

Were first isolated from thrombocytes (blood platelets).

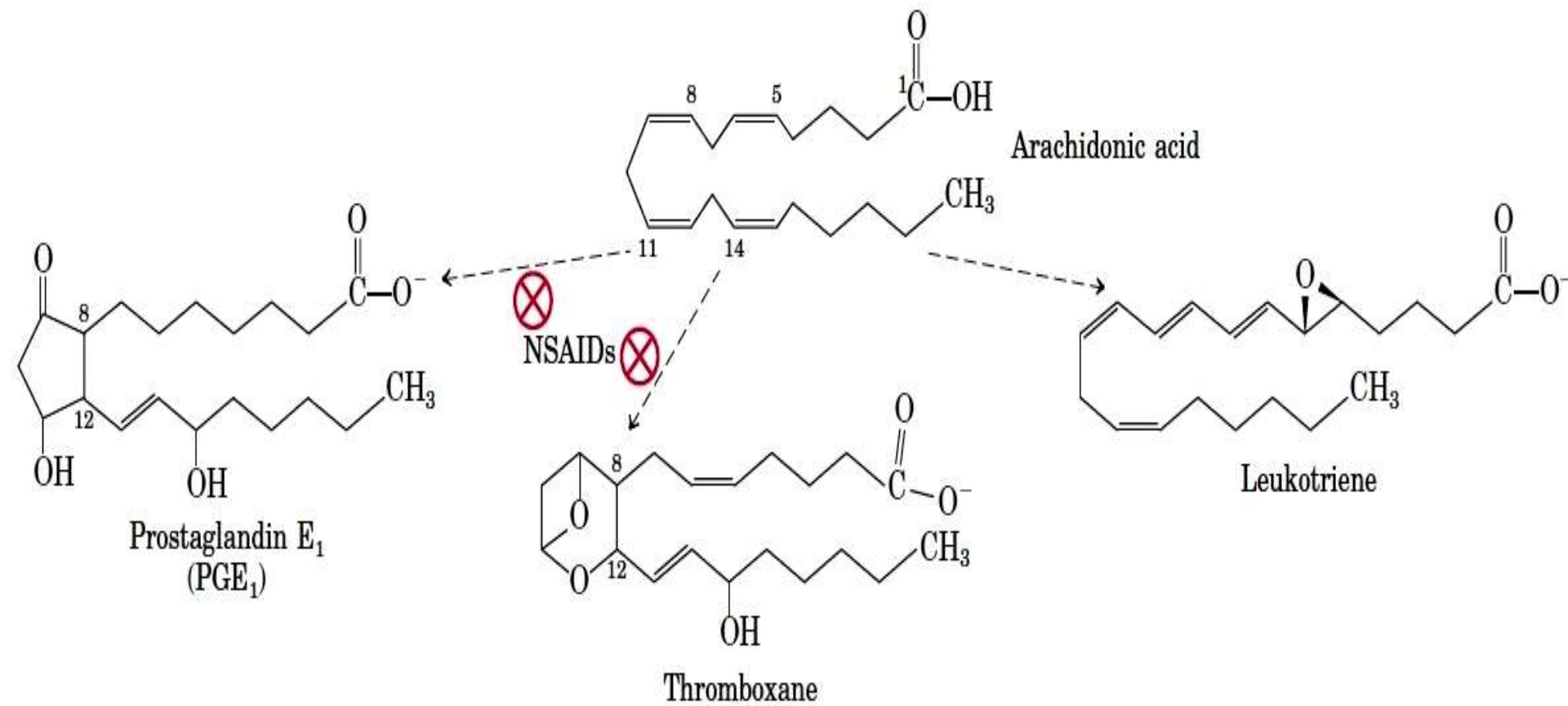
Thromboxane is known to induce platelet aggregation. When a blood vessel is ruptured, the first line of defense is the platelets circulating in the blood, which form an incipient clot. Thromboxane A₂ causes other platelets to clump, thereby increasing the size of the blood clot. Thromboxane cause constriction of blood vessels.

Leukotrienes:

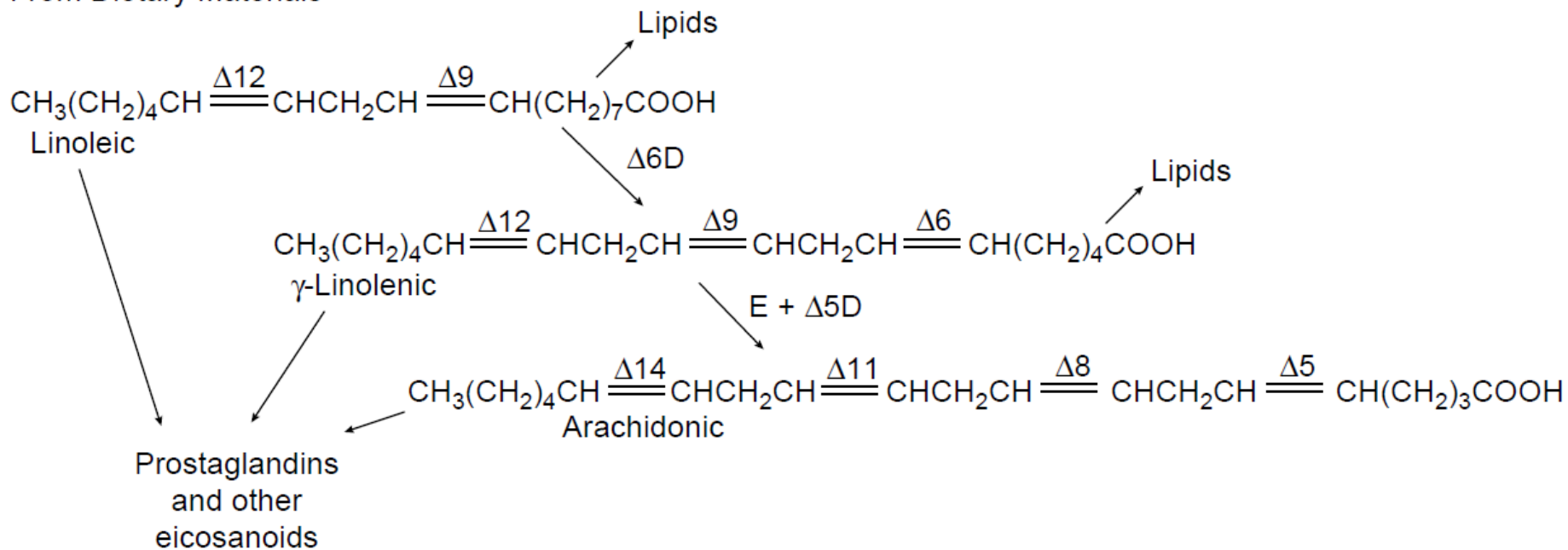
Leukotrienes is the third group of eicosanoids.

They were first found in leukocytes (white blood cells). They induce contraction of muscle lining the airways to the lungs.

Leukotrienes are potent proinflammatory agents, cause bronchoconstriction. Lipoxins are also produced via the same pathway. Overproduction of leukotrienes causes asthma-like attacks, and leukotriene synthesis is one target of antiasthmatic drugs such as prednisone.



From Dietary Materials



2- Glycerides (Triacylglycerols):

Glycerides are lipid esters that contain the glycerol molecule and fatty acid.

Triacylglycerols are not found in membranes, because they are entirely non-polar.

Triacylglycerols act as energy storage molecules, especially in adipose tissue they are also found in lipoproteins.

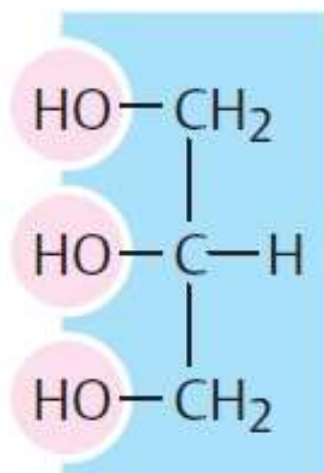
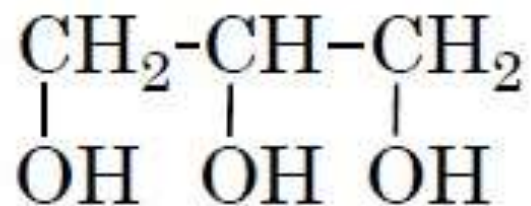
Esterification may occur at one, two or all positions of glycerol producing mono-glyceride, diglyceride and triglycerides respectively of different fatty acids, or of three identical fatty acids.

The three acyl residues of a fat molecule may differ in their chain length and the presence or absence of double bonds and number of double bonds they contain. This results in a large number of possible combinations of individual fat molecules.

Upon oxidation (Beta Oxidation) in the body, unsaturated fatty acids yield less energy than saturated ones of the same size. However, depot fat, which is an energy store, contains a high content of unsaturated fatty acids.

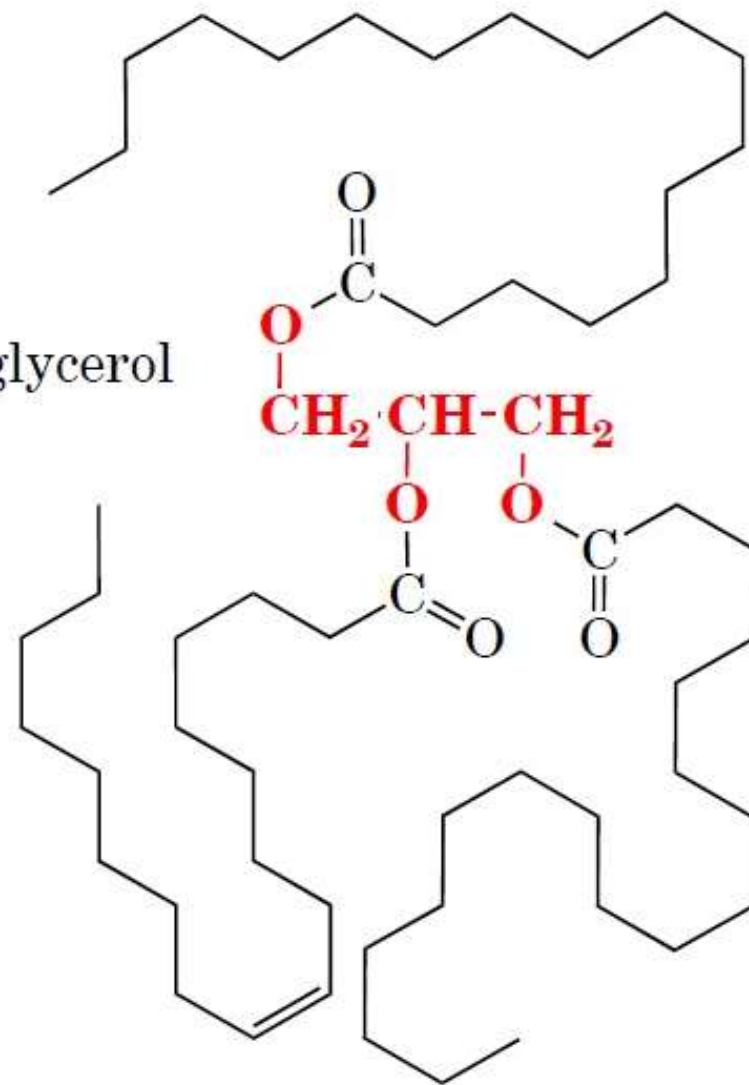
This is thought to be so because the fat must be in a liquid state to present a large surface area to enzymes that hydrolyze it. Also, solid fat would render the adipose tissue to be rigid and unyielding during mechanical stress.

Glycerol



Glycerol

Triacylglycerol



Physical Properties of Triacylglycerols

- i. They are hydrophobic and insoluble in water.
- ii. **Oils are liquids** at 20°C; they are triglycerides which contain a higher proportion of unsaturated fatty acids or short chain triglycerides. Oils are generally of plant origin.
- iii. **Fats are solids** at room temperature and contain mainly saturated long chain fatty acids. Fats are mainly of animal origin.

- iv. When the constituent fatty acids have a higher chain length and are predominantly saturated, '**hard fat**' is formed, e.g. pig fat.
- v. Fats containing medium chain triglycerides or unsaturated fatty acids are soft fats, e.g. butter, coconut oil.

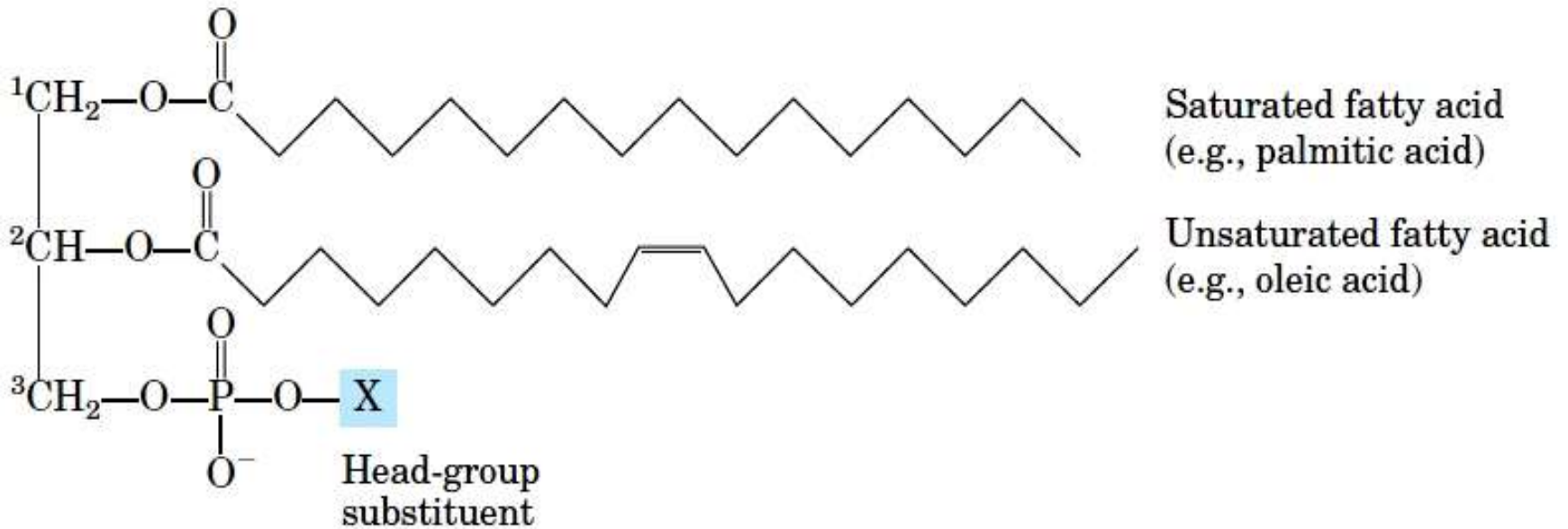
Coconut oil contains mainly medium chain TAG, e.g. Lauric and Myristic acids.

Glycerophospholipids:

Glycerophospholipids are the major component of biological membranes.

Glycerophospholipids are composed of two acyl group and a phosphate ester on the third carbon.

This makes the glycerophospholipid molecule considerably more polar, because the phosphate group is charged under physiological conditions.



Glycerophospholipid
(general structure)

Venom of vipers?

Storage of Energy as Fat

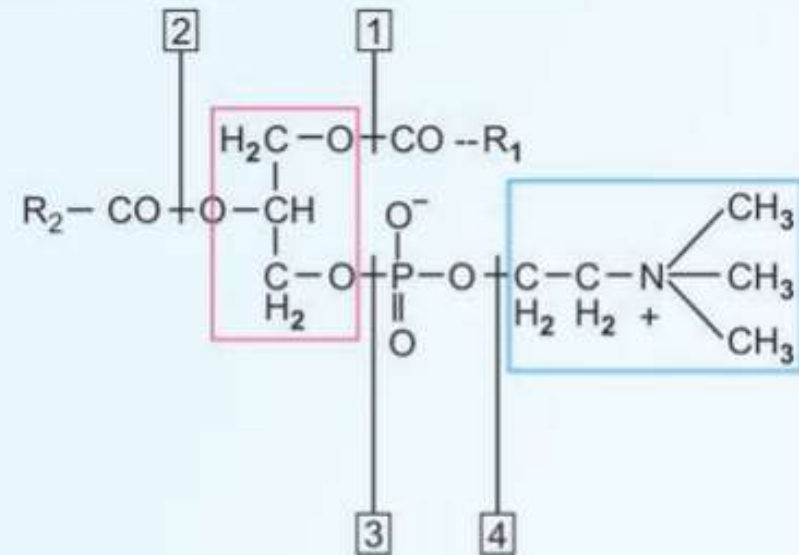
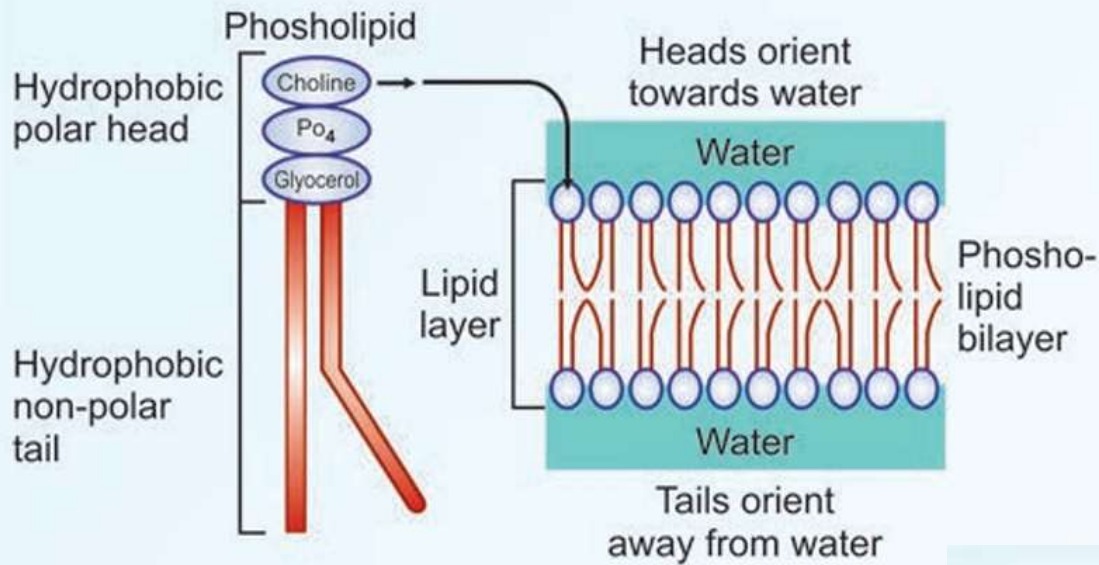
The triacylglycerols are the storage form of lipids in the adipose tissue. In a 70 kg normal person, body stores contain about 11 kg of triacylglycerol, which is roughly equivalent to 100,000 kCal. If the same calories were stored as hydrated glycogen, the total weight of this alone would have been 65 kg.

When stored as TAG, water molecules are repelled and space requirement is minimal.

Types of glycerophospholipid:-

1. Phosphatidyl choline or Lecithin

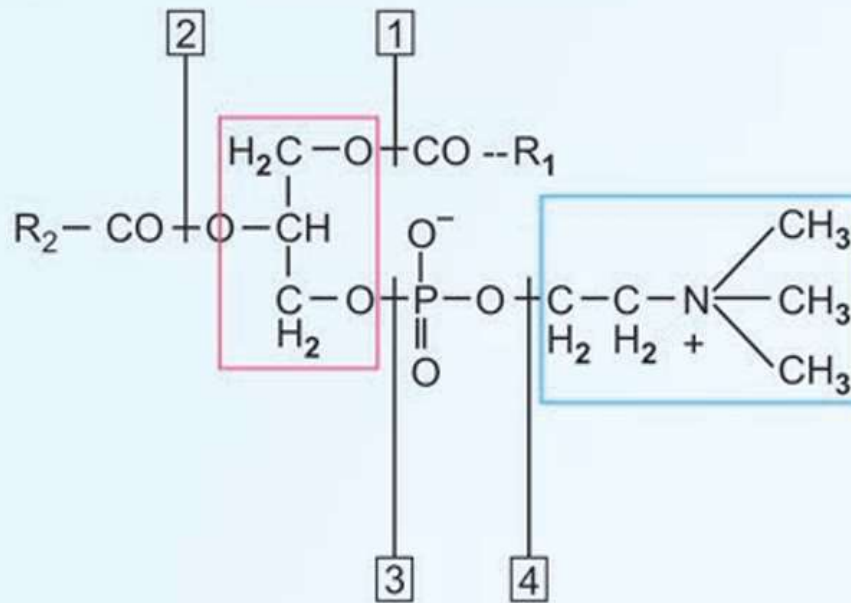
- i. This is a nitrogen containing phospholipid.
The word lecithin is derived from the Greek word, lekithos = egg yolk. It contains glycerol.
- ii. The alpha and beta positions are esterified with fatty acids. Usually the fatty acid attached to the beta-carbon is a PUFA molecule.
- iii. The phosphoric acid is added to the third position, to form phosphatidic acid. The phosphate group is esterified to the quaternary nitrogen base, Choline.



- 1 = Site of action of phospholipase A₁
- 2 = Site of action of phospholipase A₂
- 3 = Site of action of phospholipase C
- 4 = Site of action of phospholipase D

Action of Phospholipases

Phospholipases are enzymes that hydrolyse phospholipids. Different phospholipases are involved in the hydrolysis of specific bonds in lecithin. Phospholipase A₂ acts on an intact lecithin molecule hydrolysing the fatty acid esterified to the (second) carbon atom. The products are **Lysolecithin** and fatty acid. Lysolecithin is a detergent and hemolytic agent. The enzyme is present in the venom of viper snakes. So that explain hemolysis and consequent renal failure seen in viper poisoning.



1 = Site of action of phospholipase A1
 2 = Site of action of phospholipase A2
 3 = Site of action of phospholipase C
 4 = Site of action of phospholipase D

Phospholipase A2

Lecithin $\longrightarrow$ Lysolecithin + fatty acid

Phospholipase A1

Lecithin $\longrightarrow$ Acyl glycerophosphorylcholine + fatty acid

Phospholipase C

Lecithin $\longrightarrow$ 1,2 diacyl glycerol + Phosphoryl choline

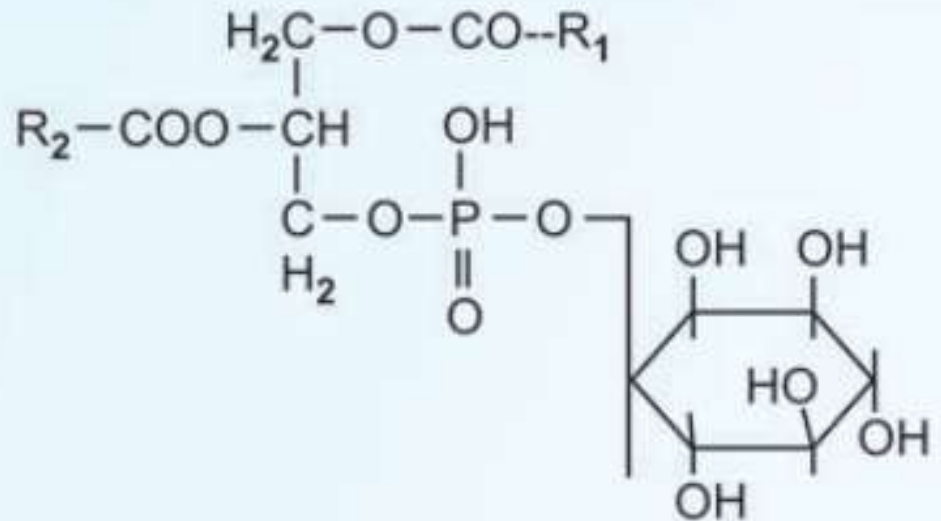
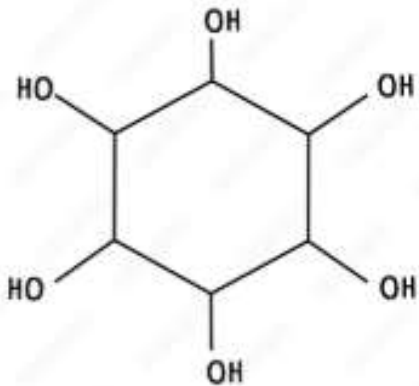
Phospholipase D

Lecithin $\longrightarrow$ Phosphatidic acid + choline

3. Phosphatidyl inositol

Here phosphatidic acid is esterified to inositol. Phosphatidyl inositol bisphosphate or PIP2 is present in biomembranes. This compound plays a vital role in the mediation of hormone action on biomembranes and acts as a second messenger.

Inositol



4. Phosphatidyl Glycerol

It is formed by esterification of phosphatidic acid to glycerol. When two molecules of phosphatidic acid are linked with a molecule of glycerol, diphosphatidyl glycerol or **cardiolipin** is formed. It is the major lipid of mitochondrial membrane.

Commercially, it is extracted from myocardium.

Decreased **cardiolipin** level leads to mitochondrial dysfunction, and is accounted for heart failure, hypothyroidism and some types of myopathies.

5- Plasmalogens

Some Phospholipids have Ether-Linked Fatty Acids instead of ester link.

In some of the glycerophospholipids, one of the two fatty acyl chains is attached to glycerol in ether, rather than, ester linkage.

The ether-linked chain may be saturated, as in the alkyl ether lipids (platelet-activating factor), or unsaturated such as plasmalogens that contain a double bond between C_1 and C_2 .

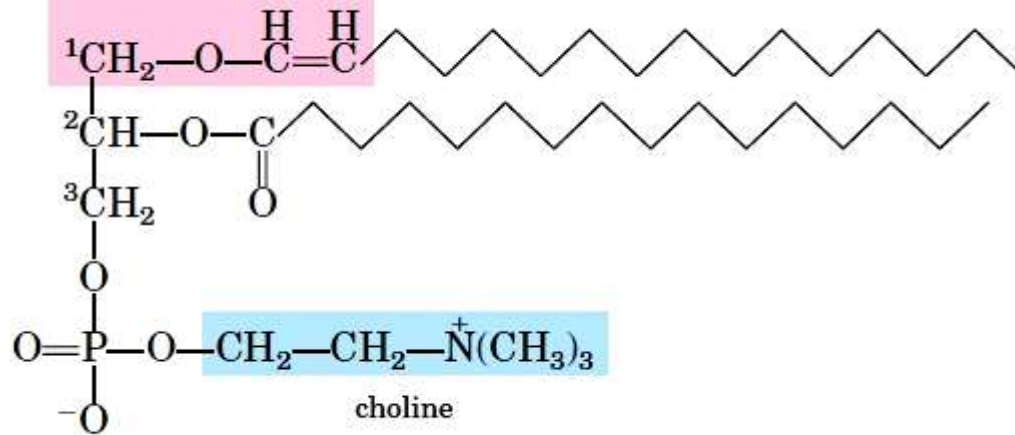
These are phospholipids which have an aliphatic long chain unsaturated alcohol in ether linkage with the first hydroxyl group of glycerol.

The second OH group is esterified to a fatty acid.

The alcohols have about C12 to C18 chain length.

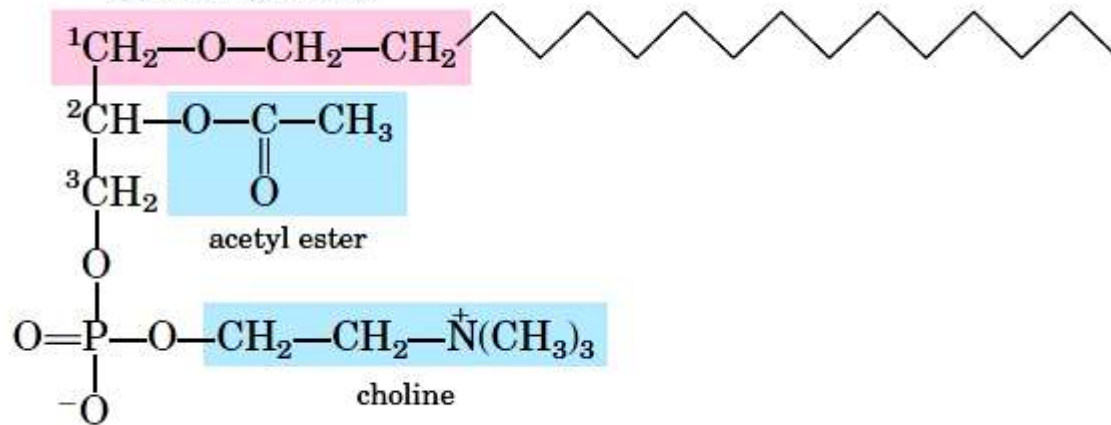
Plasmalogens are found in biomembranes in brain and heart.

ether-linked alkene



Plasmalogen

ether-linked alkane



Platelet-activating factor

Vertebrate heart tissue is uniquely enriched in ether lipids; about half of the heart phospholipids are plasmalogens.

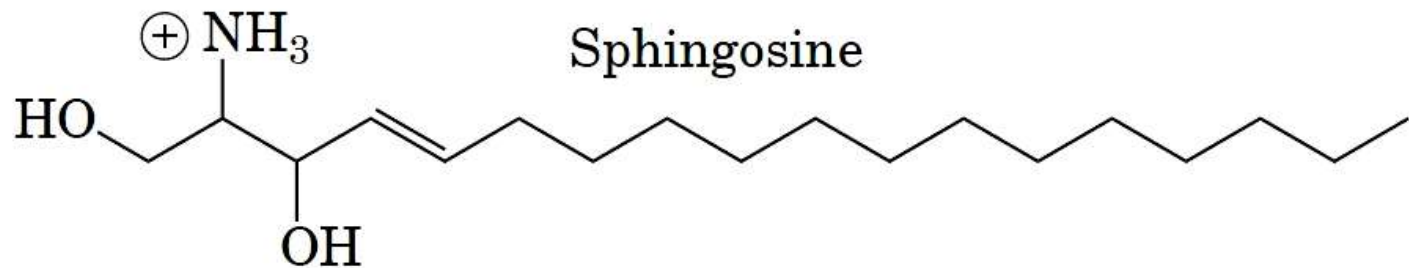
Three major classes of plasmalogens have been identified that contain ethanolamine, choline and serine as the head group.

The functional significance of ether lipids in these membranes is unknown; but they confer their resistance to the phospholipases that cleave ester-linked fatty acids from membrane lipids.

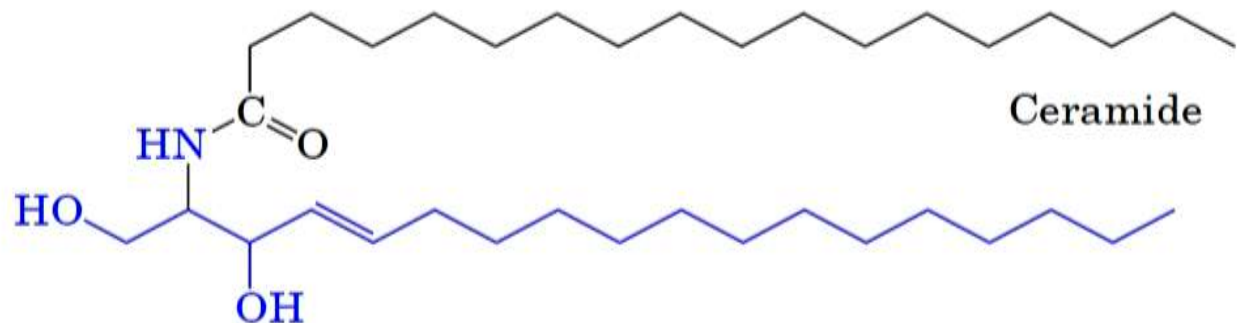
Another ether lipid is **platelet-activating factor** which released from leukocytes and stimulates platelet aggregation and the release of serotonin (a vasoconstrictor) from platelets, and plays an important role in inflammation and the allergic response.

3- Nonglyceride lipids (sphingolipids):

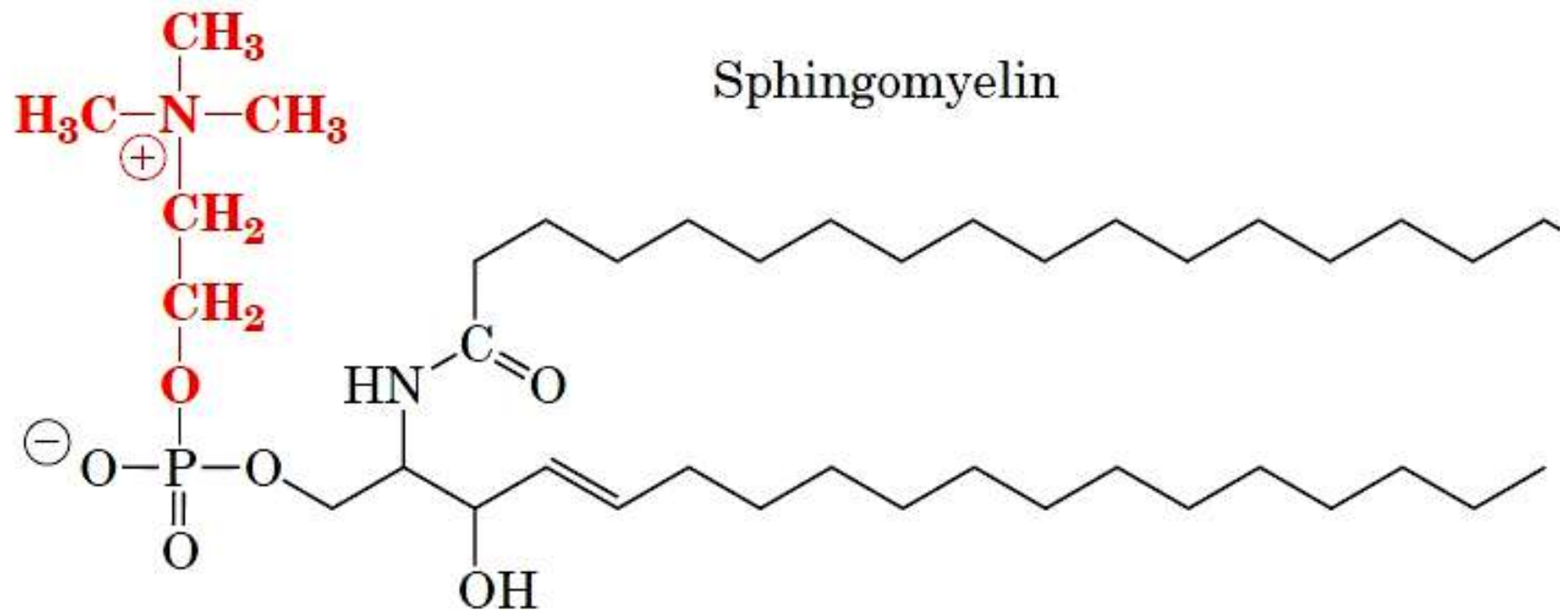
Sphingolipids are derivatives of sphingosine, an 18-carbon amino alcohol.



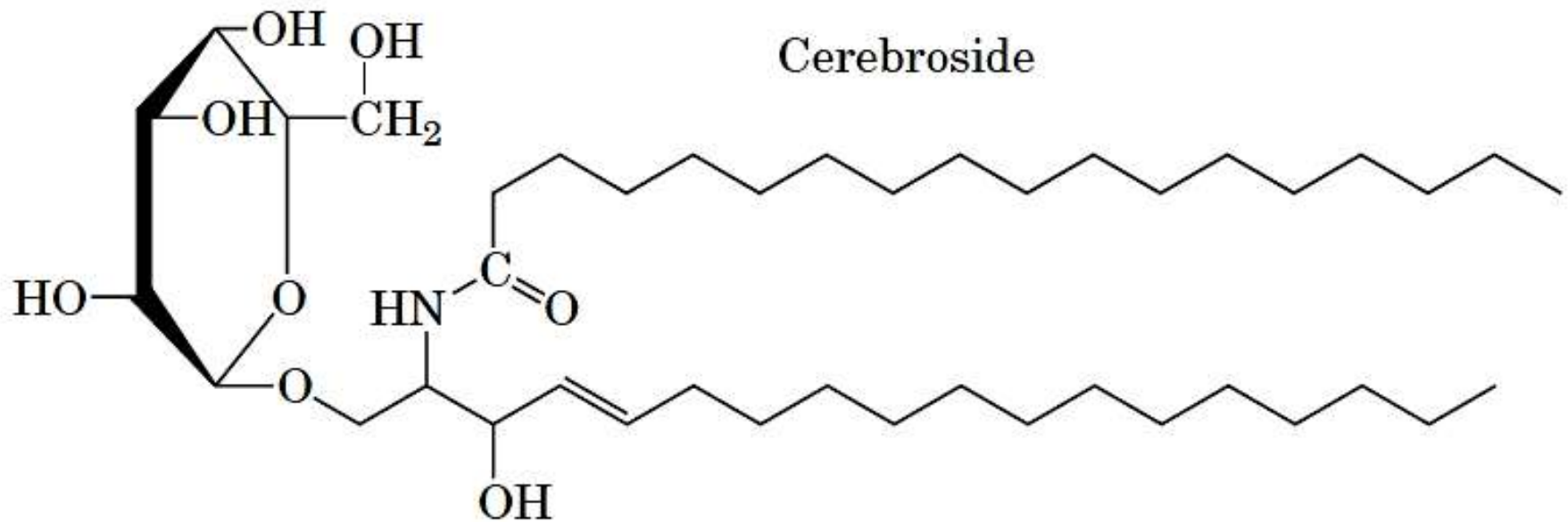
The simplest sphingolipid is **ceramide**, which is formed from sphingosine bounded in an amide linkage to a fatty acid.



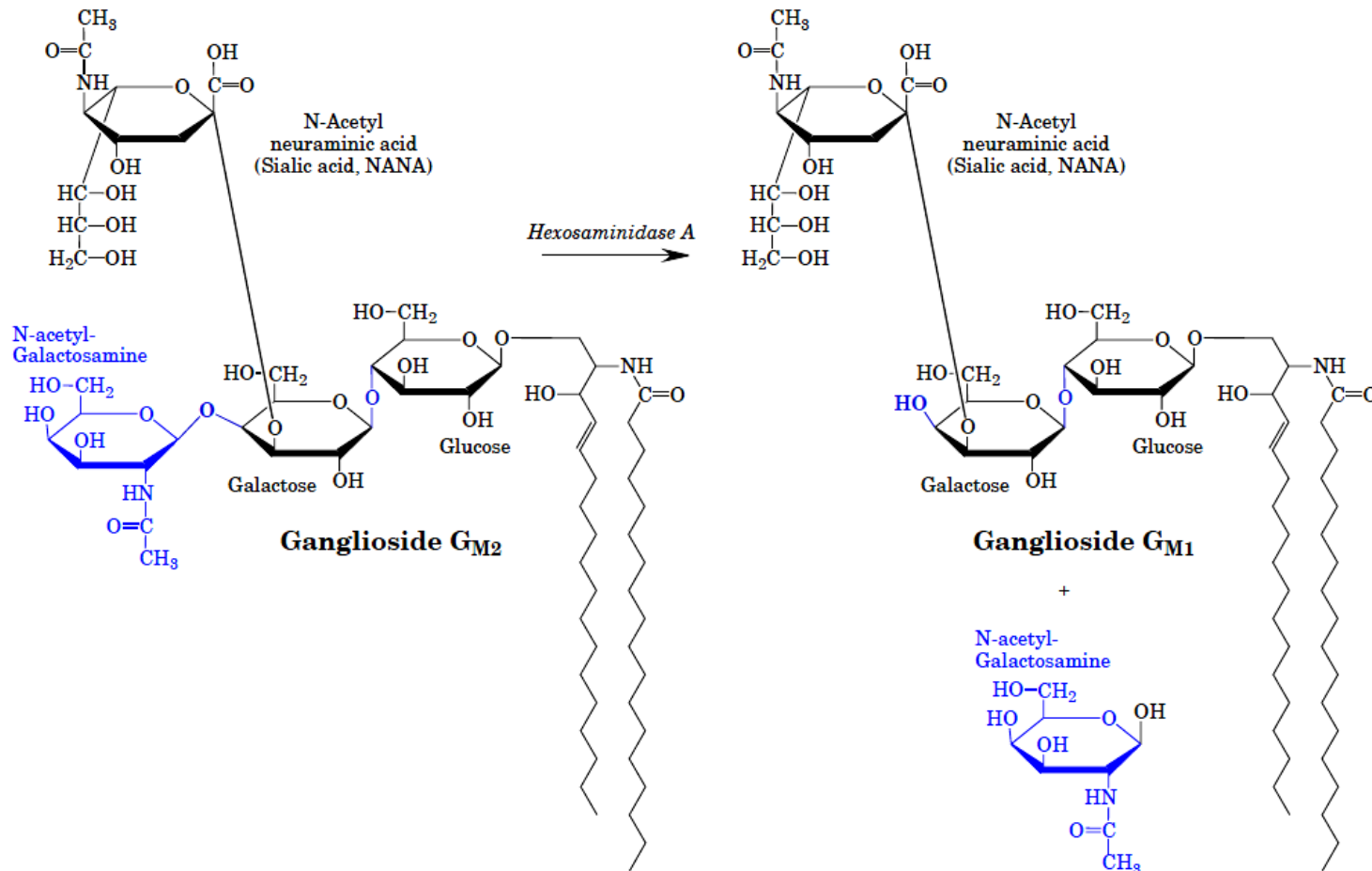
Sphingomyelins: its derived from ceramide usually have an additional group attached to the phosphate (choline or ethanolamine).



Cerebrosides are ceramide linked by an ether linkage to glucose or galactose.



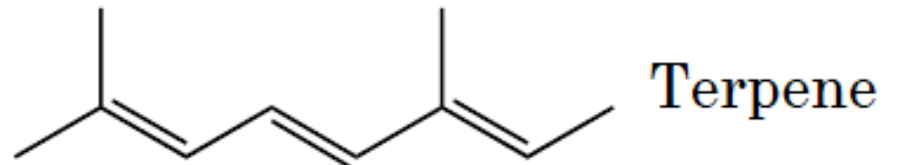
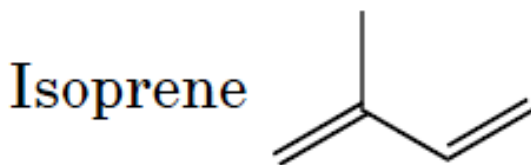
Gangliosides are related to cerebroside, but it contains three or more sugar residues.



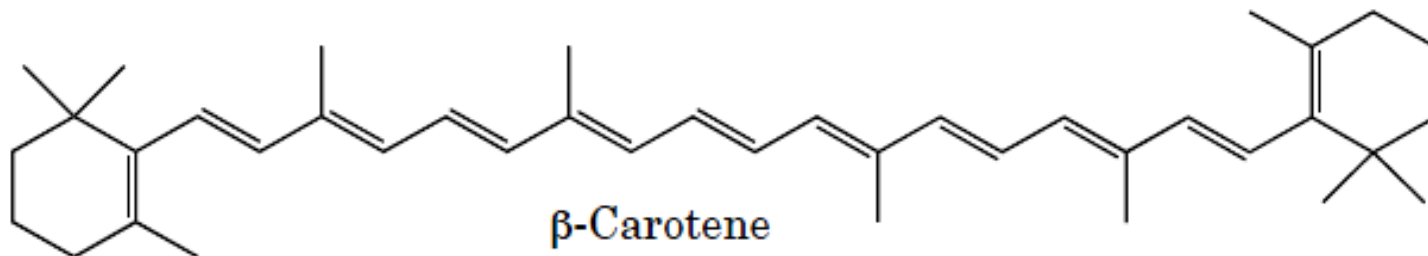
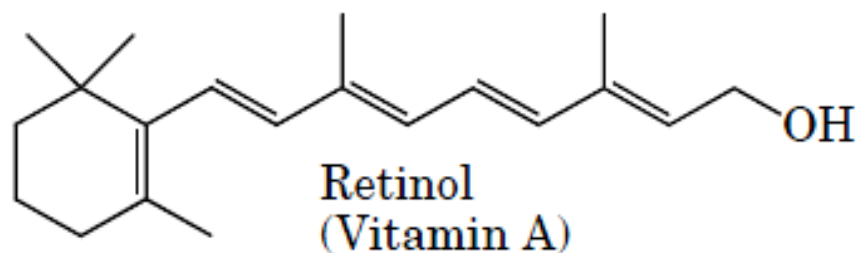
Steroids:

The steroids are members of a large collection of lipid called the isoprenoids.

Terpene is composed of two isoprene molecules.

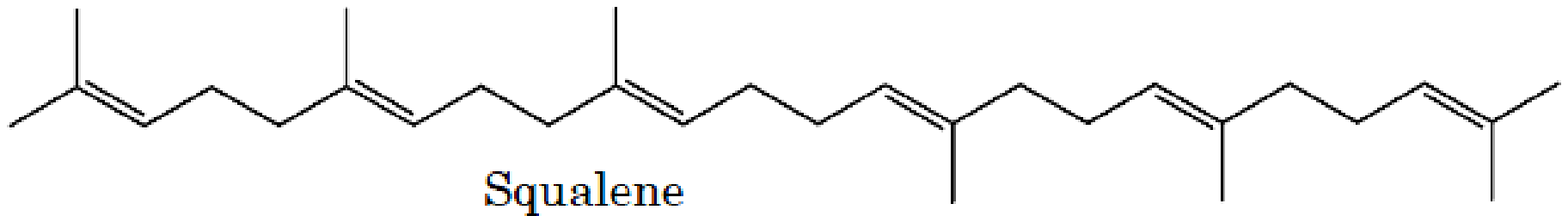


Examples of terpenes include the **diterpene** retinol (vitamin A) and the plant compound β -carotene, a **tetraterpene** that can act as a precursor of retinol.



Squalene is a **triterpene** (30 carbons) compound it acts as a precursor of cholesterol.

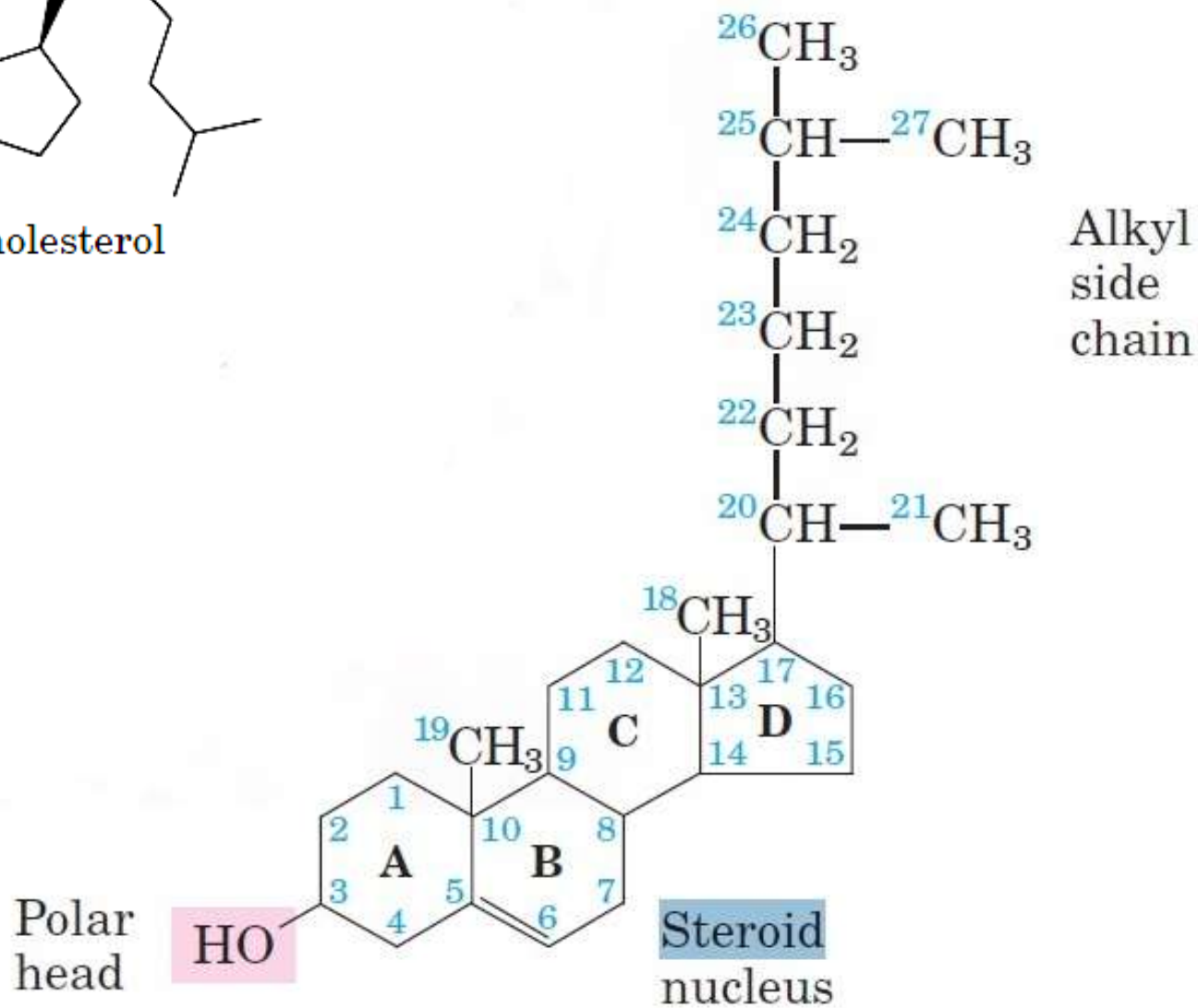
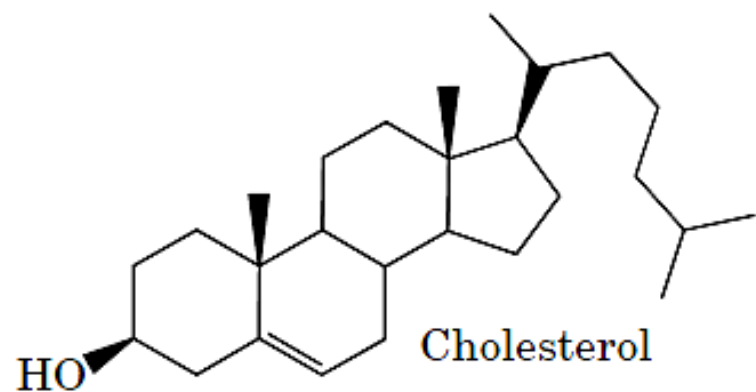
Steroids are compounds containing the steroid nucleus, which consists of 3 cyclohexane rings and one cyclopentane ring fused together.



Cholesterol is a 27- carbon terpene derivative.

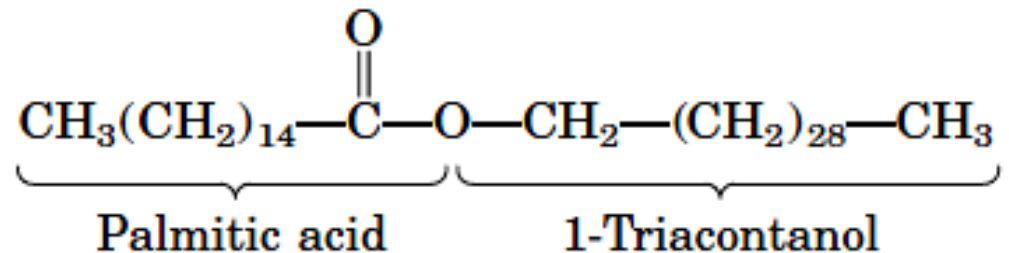
Most of the cholesterol are incorporated into the lipid layer of plasma membranes, or converted into bile acids. A very small amount of cholesterol is used for biosynthesis of the steroid hormones.

Cholesterol is amphipathic, with a polar head group (the hydroxyl group at C₃) and a nonpolar hydrocarbon body (the steroid nucleus and the hydrocarbon side chain).



Waxes

Waxes are esters of long chain (14 to 36 carbon) saturated or unsaturated fatty acids with long chain (16 to 30 carbon) alcohols.



Importance of wax :

1- Skin glands of certain vertebrates secrete waxes to protect their hair and skin and keep them lubricated and waterproof.

- 2- Certain birds have wax secreting glands to prevent their feather from being wet.
- 3- The leaves of certain tropical plants are also coated with wax to prevent excessive evaporation of water and for protection against parasites.

4- Complex lipids: Lipids which are bonded to other type of molecules.

The most important complex lipids are plasma lipoproteins which are responsible for the transport of other lipids in the body.

Lipoprotein particles consist of a core of hydrophobic lipids (cholesterol ester or TG) surrounded by amphipathic proteins (containing both hydrophobic and hydrophilic region), phospholipids and cholesterol.

Chylomicrons:

Carrying dietary TG from the intestine to other tissue.

Very low density lipoprotein (VLDL):

VLDL is synthesized and released by the liver. VLDL is used to transport triacylglycerol from the liver to other tissues.

Low density lipoprotein (LDL):

They carry cholesterol to peripheral tissue and helps to regulate cholesterol in those tissues.

High levels of LDL cholesterol are associated with elevated risk of heart disease. LDL cholesterol is called the “bad cholesterol”.

High density lipoprotein (HDL):

They are bound to plasma cholesterol; however, they transport cholesterol from peripheral tissue to the liver.

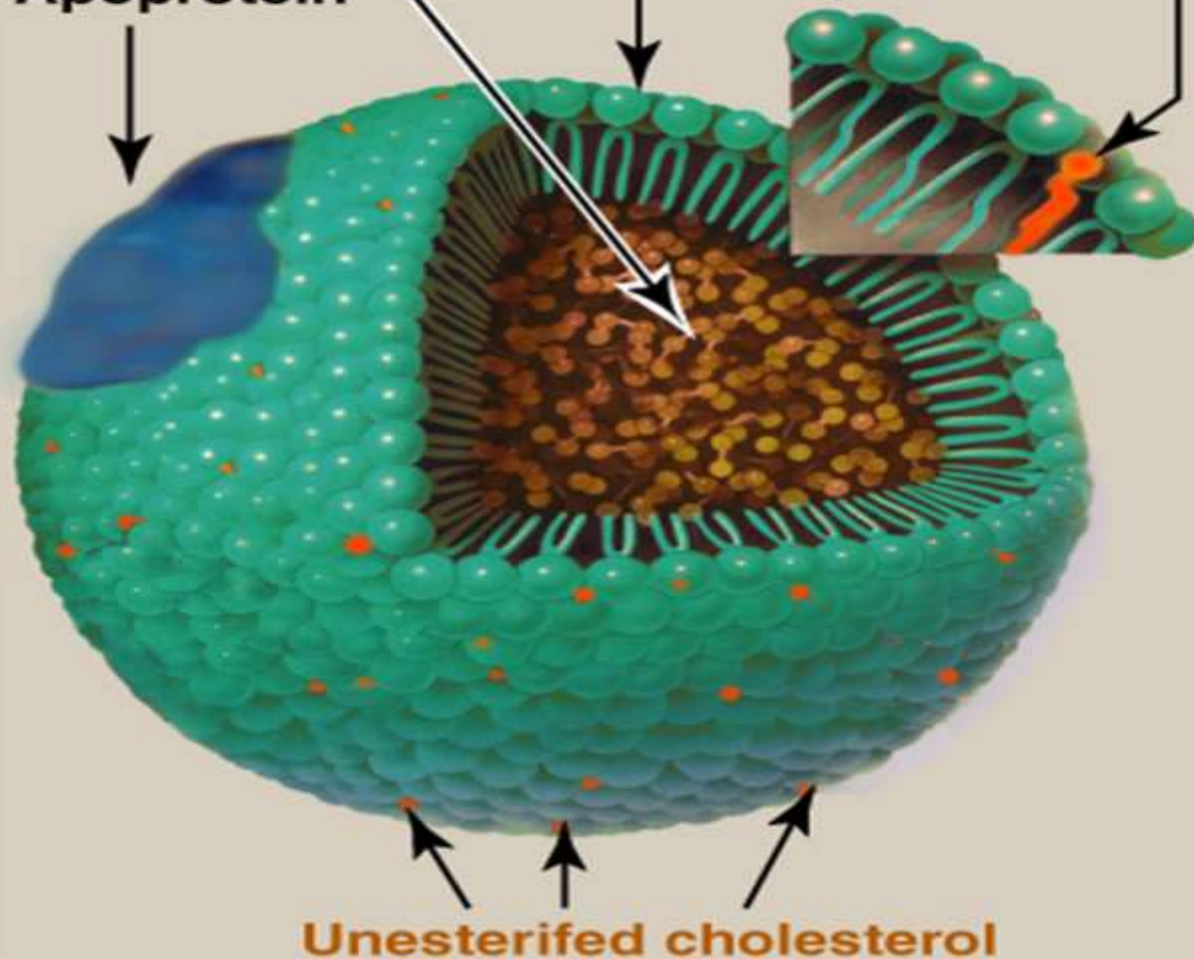
High levels of HDL are associated with reduced risk of heart disease, possibly due to increased cholesterol scavenging by HDL, and therefore lower LDL and total plasma cholesterol levels. HDL cholesterol is called the “good cholesterol”.

**Inner core of
triacylglycerols
and cholesteryl
esters**

Phospholipids

**Unesterified
cholesterol**

Apoprotein



Rancidity of Fats:

The unpleasant odor and taste developed by most natural fats upon ageing is referred to as rancidity.

Rancidity may be due to hydrolysis of component glycerides of a fat into free fatty acids and glycerol.

If the hydrolysis is incomplete, a mixture of glycerol, monoglycerides and diglycerides with free fatty acids may be produced.

Rancidity may also be caused by various oxidative processes. For example, oxidation at the double bonds of unsaturated fatty acyl residues may form peroxides, which then decompose to form aldehydes of bad odor and taste. This process is greatly increased by exposure to light.

Addition of minute amounts of antioxidants can prevent the oxidation and rancidity.

Biological advantage of Fats

Triacylglycerols are ideal molecules for efficient storage of metabolic energy. This is because they are more highly reduced than the other molecules such as carbohydrates or proteins and hence yield significantly more energy upon oxidation.

Another advantage is that, fats being nonpolar substances, can be stored in anhydrous form without carrying an extra load of water and occupying less space during storage.

Glycogen present in human body can meet the metabolic energy requirement only for less than a day, whereas the fat stored in adipocytes can fuel the mammals for days and even weeks.

The fat content of normal men and women is about 20% and 26% respectively, which allows them to survive for several days of starvation.

In hibernating animals, huge fat accumulated before hibernation serves dual purpose of energy storage and insulation.

Fat is also stored in the form of oil in seeds of many plants providing energy and biosynthetic precursors during seed germination. Both adipocytes and germinating seeds contain lipases that catalyze the hydrolysis of stored triacylglycerols, releasing fatty acids for export to sites where they are required as fuel.

Clinical Applications

1. Excessive fat deposits cause obesity which is a risk factor for heart attack.
2. Abnormality in cholesterol and lipoprotein metabolism leads to atherosclerosis and cardiovascular diseases.
3. In diabetes mellitus, the metabolisms of fatty acids and lipoproteins are changed, leading to ketosis.